arXiv:2506.08572v2 [cs.LG] 4 Jul 2025

The Geometries of Truth Are Orthogonal Across Tasks

Waiss Azizian [1] **Michael Kirchhof** [1] **Eugene Ndiaye** [1] **Louis Bethune** [1] **Michal Klein** [1] **Pierre Ablin** [1] **Marco Cuturi** [1]

Abstract

Large Language Models (LLMs) have demonstrated impressive generalization capabilities across various tasks, but their claim to practical relevance is still mired by concerns on their reliability. Recent works have proposed examining the activations produced by an LLM at inference time to assess whether its answer to a question is correct. Some works claim that a "geometry of truth" can be learned from examples, in the sense that the activations that generate correct answers can be distinguished from those leading to mistakes with a linear classifier. In this work, we underline a limitation of these approaches: we observe that these "geometries of truth" are intrinsically task-dependent and fail to transfer across tasks. More precisely, we show that linear classifiers trained across distinct tasks share little similarity and, when trained with sparsity-enforcing regularizers, have almost disjoint supports. We show that more sophisticated approaches (e.g., using mixtures of probes and tasks) fail to overcome this limitation, likely because activation vectors commonly used to classify answers form clearly separated clusters when examined across tasks.

1. Introduction

Large Language Models (LLMs) have seen tremendous success in recent years across a wide range of tasks. However, their widespread deployment is not without risks: from hallucinations (Ji et al., 2023) to outright deception (Park et al., 2024), the complexities underpinning LLM generation can be the root of many issues. These challenges are particularly concerning in high-stakes domains like healthcare, legal advice, and financial analysis, where incorrect or misleading information can lead to serious harm. As a result, several works have suggested leveraging the various activations generated by a model at inference time to understand and assess the truthfulness of its output. Azaria & Mitchell (2023) demonstrated that training a simple classifier on top of the hidden activations of LLMs can help predict whether an LLM has provided a truthful answer to a user-provided question. This finding suggests that models somewhat reveal enough information in their activations at inference time to help users assess whether they are producing correct information. Numerous subsequent works have explored this "geometry of truth" and confirmed that it is approximately linear, in the sense that a linear classifier can distinguish reliably truthful from erroneous answers (Li et al., 2023; Marks & Tegmark, 2024; Xiong et al., 2024; Burns et al., 2023; Kossen et al., 2025). As a token of their relevance, these directions can then be used to steer the LLM towards factual generations (Li et al., 2023; Wang et al., 2024), and are increasingly studied as a cheap and effective proxy to assess the uncertainty of an LLM output on a given task, e.g. (Sky et al., 2024; Zhang et al., 2025; Slobodkin et al., 2023).

But is one hyperplane all it takes? While the literature abounds with examples of the relevance of such probes when tested within a given task or knowledge domain, it remains unclear whether they can generalize across different tasks. Although some works have reported encouraging results in that direction (Azaria & Mitchell, 2023; Marks & Tegmark, 2024; Beigi et al., 2024), others provide a mixed assessment (Slobodkin et al., 2023; Kossen et al., 2025; Zhang et al., 2025), while some works (Orgad et al., 2025; Levinstein & Herrmann, 2024; Sky et al., 2024) show on the contrary that probes completely fail to generalize in some settings. In that context, one might be tempted to increase the complexity of probes and their training procedures, such as the more sophisticated pipeline proposed by Beigi et al. (2024) that incorporates augmentations to build additional representations on top of the activations of multiple layers taken jointly. Taken together, these works paint a mixed picture of whether simple probes can generalize at all, and if they do not, whether their disappointing results are an artifact of their training or an intrinsic limitation.

Our contributions. Our work proposes to answer whether there is any hope to see linear probes transfer reliably *across tasks*. We study the variability of these probes across tasks

[1]Apple. Correspondence to: Michael Kirchhof, Marco Cuturi <contact see website>.

Published at ICML 2025 Workshop on Reliable and Responsible Foundation Models. Copyright 2025 by the author(s).

and reach the following findings:

Task-specific truthfulness geometries. We first demonstrate that truthfulness geometries are fundamentally task-specific. Through comprehensive cross-task evaluation on seven diverse datasets, we show that linear probes trained on different tasks exhibit distinct "geometries of truth" that fail to generalize. While some task pairs show successful transfer, most combinations result in substantial performance degradation. This systematic analysis shows that generalization success depends critically on task similarity rather than being a universal property of truthfulness probes.

Geometric analysis of orthogonality. We then provide geometric analysis revealing why these failures occur. We demonstrate that truthfulness directions are largely orthogonal across tasks, with a clear correlation between geometric similarity and generalization performance. Using sparse probes, we reveal that probe supports are nearly disjoint across tasks, providing interpretable evidence for orthogonality. Visualizations show that different tasks form distinct clusters in the representation space of the model, confirming our geometric explanation for the failure to transfer.

Orthogonality persists in multi-task settings. Finally, we test whether our orthogonality hypothesis holds when training on mixtures of task. We demonstrate that training on diverse task mixtures fails to resolve generalization problems, and critically show that optimal directions for target tasks cannot be recovered through linear combinations of directions from other tasks. We further show that more complex architectures are also no better than naive parameter summation, suggesting the limitation is intrinsic. Given these findings, we explore conservative deployment strategies using conformal prediction to maintain reliability guarantees in cross-task scenarios.

Our work is structured as follows:

- In Section 2, we review the necessary background on probing and present a detailed survey of the literature on generalization properties of uncertainty probes.
- In Section 3, we introduce our experimental setup (models and datasets) and systematically study cross-task generalization failures through geometric analysis of probe directions and sparse probe supports.
- In Section 4, we examine whether training on mixtures of tasks can overcome these limitations, test more complex architectures, and explore conservative deployment strategies.

2. Background

Linear probing for uncertainty quantification. Uncertainty quantification for LLMs has attracted significant attention recently. Apart from relatively more costly multi-samples methods such as semantic entropy (Farquhar et al., 2024), many efforts have focused on learning simple classifiers whose inputs are activation vectors computed by the LLM at inference time, using labeled pairs of questions and either correct or incorrect answers. Azaria & Mitchell (2023) introduced this approach for uncertainty quantification for LLMs. While the classifier was originally set to be a multi-layer perceptron, follow-up works showed that using a simple linear logistic regression model could achieve similar performance (Li et al., 2023; Orgad et al., 2025; Marks & Tegmark, 2024; Santilli et al., 2025). In other words, and following Marks & Tegmark's terminology, there might be a linear "geometry of truth" that can separates the representations of correct from incorrect outputs.

The Geometry of Truth Hypothesis. Given a user question q, the LLM generates an answer $\hat{a} = (\hat{a}_1, \dots, \hat{a}_T)$ autoregressively. At each step $t \in [T]$, the model produces hidden state vectors $h_{t,\ell} \in \mathbb{R}^d$ where ℓ indexes the layer and t the token position. In this work, we focus on a fixed layer $\ell^\star$ (e.g. 28 and 21 for Qwen models) and extract the representation $h_{T,\ell^\star}$ at the final token of the output (or the token before, $T-1$). To provide supervision, we follow (Farquhar et al., 2024; Santilli et al., 2025) and label the correctness of $\hat{a}$ given the gold answer a using LLM-as-a-judge (Zheng et al., 2023). This yields a label $y \in \{-1, +1\}$ which is positive is the answer $\hat{a}$ is correct and negative otherwise. This provides a dataset $\mathcal{D} = \{(h_i, y_i)\}_{i \in [N]}$ where each point $h_i = h_{T,\ell^\star}$ is the final hidden representation for a generated answer i. It is the internal state right before the model decides what the final answer token will be. We then train a linear probe i.e. a logistic regression classifier to predict correctness from the hidden states:

$$\min_{\substack{\theta \in \mathbb{R}^d, \\ b \in \mathbb{R}}} \frac{1}{N} \sum_{i=1}^{N} \log\left(1 + e^{-y_i(\theta^\top h_i + b)}\right) + \frac{\lambda_2}{2} \|\theta\|_2^2 \qquad (1)$$

The *geometry of truth hypothesis* states that truthful and untruthful generations are linearly separable in model's hidden space, such that a single hyperplane parametrized by (θ, b) can distinguish correct from incorrect outputs. We explore if this is universal or task-specific.

On the Generalization of Uncertainty Probes. The generalization of uncertainty estimates has been studied in several works but there is still no definitive consensus, neither among papers nor within the papers themselves, as many of them conclude with a nuanced assessment. Some of the historically first results were promising:

- Azaria & Mitchell (2023) introduce their own dataset of facts, the *true-false* dataset, made of several splits with different subjects. With their carefully constructed dataset, they show that probes trained on several subjects can accurately predict the correctness of facts of another subject.

- Kapoor et al. (2024) consider finetuning the whole LLM to obtain better uncertainty probes; although they do not conduct systematic evaluations, they report good performance on a couple of datasets that were not seen during training.

Other works apply probes in slightly different contexts and report mixed generalization results.

- Slobodkin et al. (2023) leverage probes to classify unanswerable queries and study their generalization performance on datasets not seen at training (their Figure 6). Although performance decreases, the authors still note that it remains better than probes trained on the first hidden layer on the correct dataset.
- Zhang et al. (2025) use uncertainty probes for reasoning tasks and find that, although these probes generalize across similar datasets, they fail when the type of reasoning required changes.
- Kossen et al. (2025) suggest training probes with semantic entropy (Farquhar et al., 2024) and show that, for generalization to new tasks, this improves robustness in the choice of layer (their Figure 6).
- Kadavath et al. (2022) introduces a finetuning approach trained on a diverse mixture of tasks which is evaluated both on a held-out dataset and when training on only one task: the authors observe a decrease in performance but still note decent generalization.

Then there are paper that report generally negative generalization results.

- Beigi et al. (2024) report significantly worse performance on out-of-distribution data (Table 3) despite using all hidden states of an LLM as an input for an uncertainty estimator.
- Marks & Tegmark (2024) study, in a controlled setting similar to (Azaria & Mitchell, 2023), the geometry of representations of correct and incorrect answers as well as generalization properties of probes across datasets. Though the authors note that training on datasets and their negations helps, the performance of probes on out-of-distribution data remains suboptimal (Figure 5). They also visualize the hidden spaces at different layers to understand when a linear representation of uncertainty emerges.
- Levinstein & Herrmann (2024) reproduce the setting of Azaria & Mitchell (2023) and note that their probes catastrophically fail to generalize under trivial changes like introducing negations (§4.4), with a roughly 20% accuracy loss.
- In Orgad et al. (2025), the authors take a systematic approach: they consider a wide variety of question-answers datasets and find that probes trained on each of these datasets fail to generalize (their Figure 3).
- Sky et al. (2024) consider ensembles of attention-based probes for hallucination detection but notes that they do not generalize, both when trained on one dataset and tested in another (their Table 6) and when trained in two tasks and tested in a third (Table 8). This work raises the question of whether having a larger mixture of tasks night help, which we will answer by the negative in this work.

3. Truthfulness directions across tasks

After having reviewed the necessary background and the literature on the generalization of probes, we now present our experimental setup before starting our study of cross-task generalization.

Training Probes. To obtain uncertainty estimates with linear probes, we follow standard practices (Li et al., 2023; Orgad et al., 2025; Marks & Tegmark, 2024; Santilli et al., 2025) and train the probes using the logistic regression implementation of Pedregosa et al. (2011). These probes are trained with L_2 regularization with the hyperparameter tuned with cross-validation on the training set. We consider two standard token positions t: the stop token of the output and the token before the stop token of the output. Similarly, we base the probes on two embeddings: those of the last hidden layer and those at 75% depth.

Models. We consider the following models: Qwen 2.5 7B Instruct (Qwen authors, 2024), Phi 4 Mini (Phi-4 authors, 2024) and Llama 3.1 8B Instruct (The Llama 3 authors, 2024). In the main text, we consider probes trained via Equation (1) that operate on the stop token at the last layer (layer 28) of Qwen 2.5 7B Instruct. All other combinations give similar results, which we report in the appendix.

Datasets. To study generalization, we use several datasets that are variously related: NQ (Kwiatkowski et al., 2019) and SimpleQA (Wei et al., 2024) are general question-answering datasets. TriviaQA (Joshi et al., 2017), and SQUAD (Rajpurkar et al., 2016) are reading-comprehension datasets with a context, but also on generic topics. BioASQ (Nentidis et al., 2023) is composed of biology questions. SVAMP (Patel et al., 2021), made of simple arithmetic questions, and GSM8K (Cobbe et al., 2021) of more complex math word problems.

Cross-task generalization failures. We train linear probes for each dataset, then systematically evaluate cross-task transfer performance. Our dataset collection enables analysis of both successful and failed generalization cases. Figure 1 confirms widespread generalization failures, in line with Orgad et al. (2025), but reveals a crucial pattern: tasks cluster into semantically coherent groups. Factual recall tasks (TriviaQA, NQ, SimpleQA) show mutual transferability, while specialized domains (BioASQ biol-